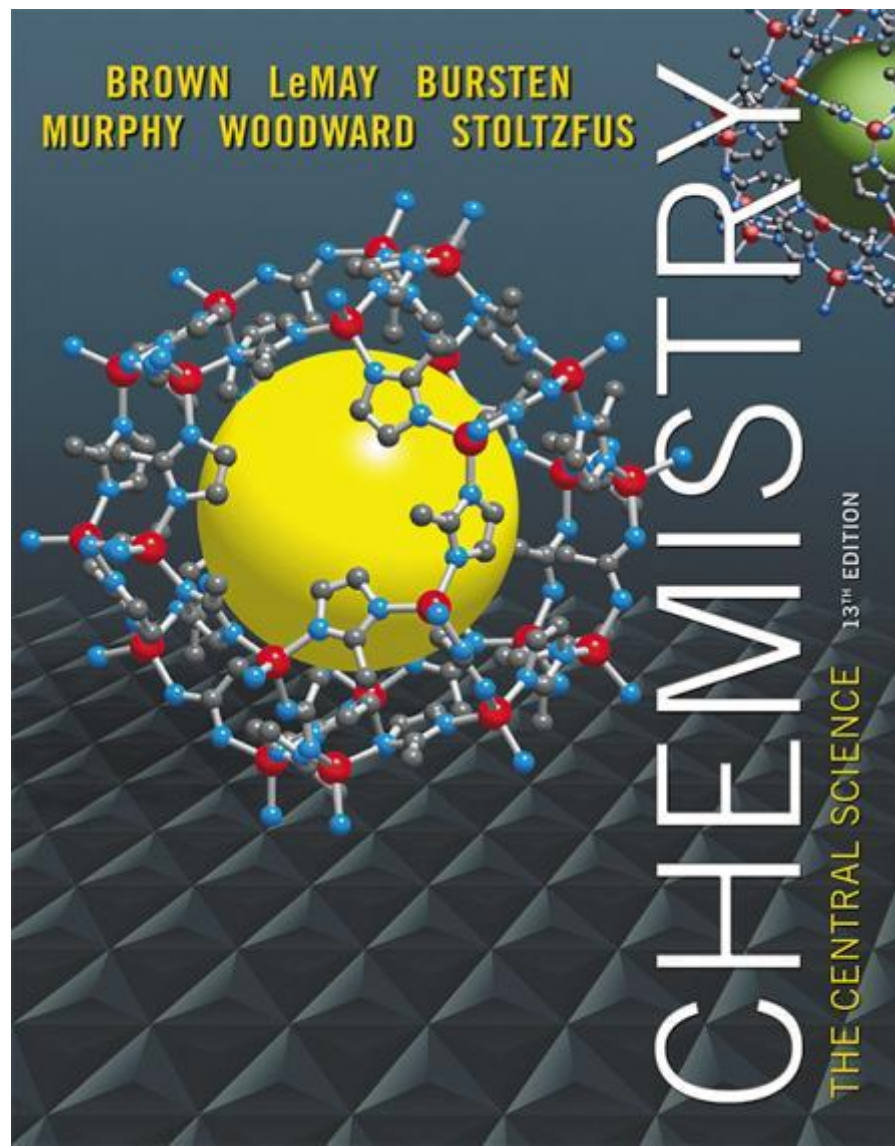


Chapter 4. Acid-Base Equilibria



Reference:

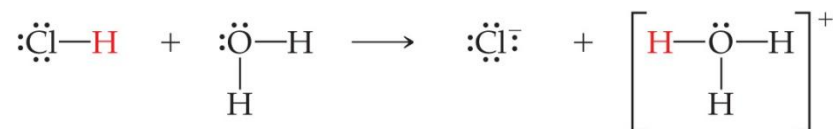
Chapter 16

Acid–Base Equilibria

Brønsted–Lowry Acid and Base

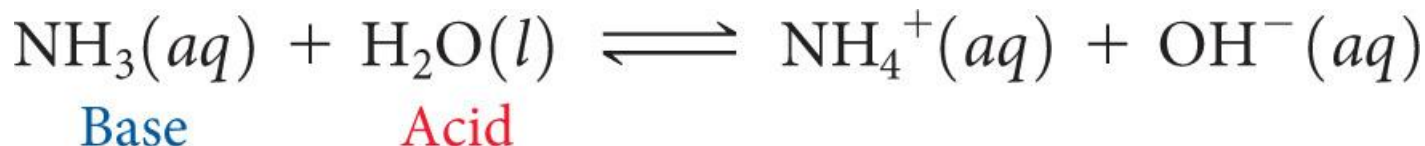
A Brønsted–Lowry **acid** must have at least one removable (acidic) proton (H^+) to donate.

A Brønsted–Lowry base must have at least one nonbonding pair of electrons to accept a proton (H^+).



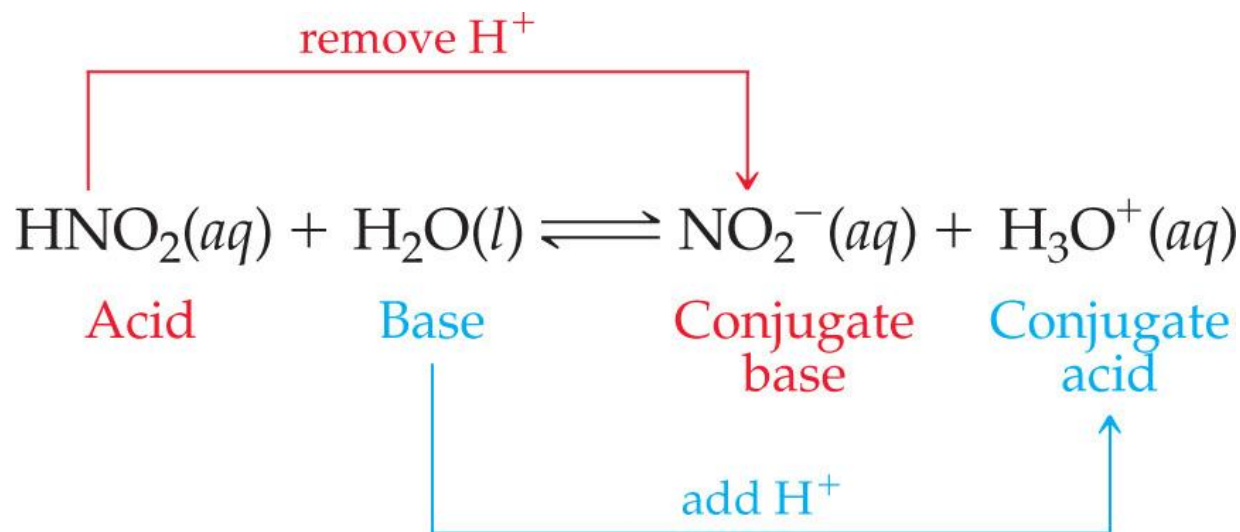
What Is Different about Water?

- Water can act as a Brønsted–Lowry base and accept a proton (H^+) from an acid, as on the previous slide.
- It can also donate a proton and act as an acid, as is seen below.
- This makes water **amphiprotic**.



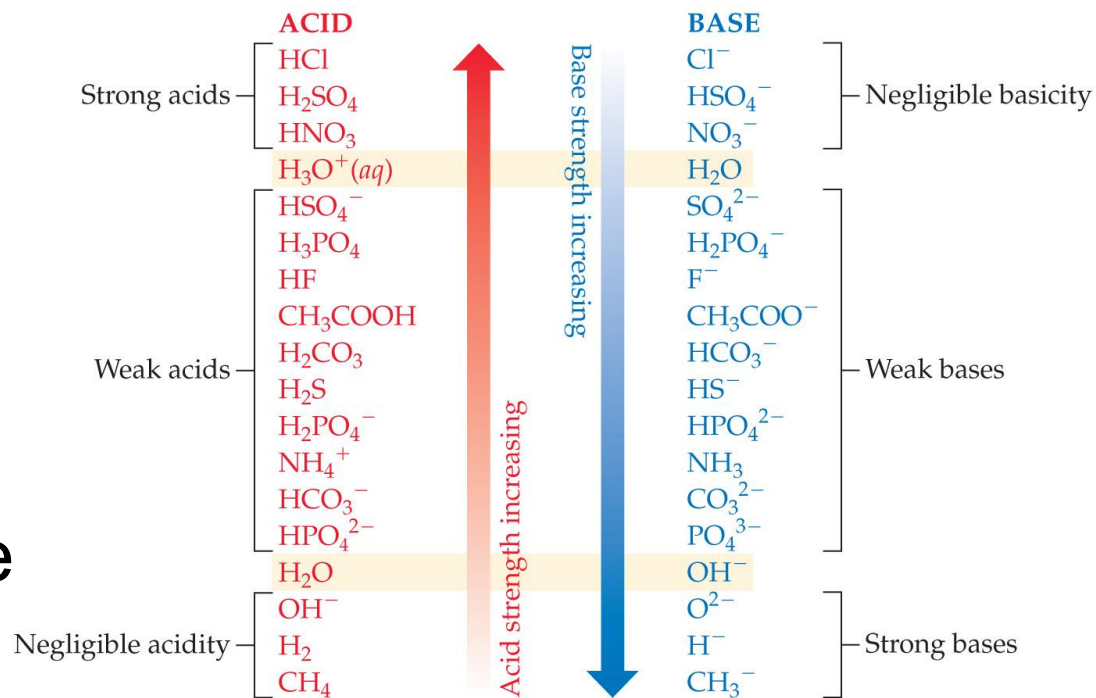
Conjugate Acids and Bases

- The term **conjugate** means “joined together as a pair.”
- Reactions between acids and bases always yield their conjugate bases and acids.



Relative Strengths of Acids and Bases

- Acids above the line with H_2O as a base are *strong acids*; their conjugate bases do *not* act as **bases** in water.
- Bases below the line with H_2O as an acid are *strong bases*; their conjugate acids do *not* act as acids in water.



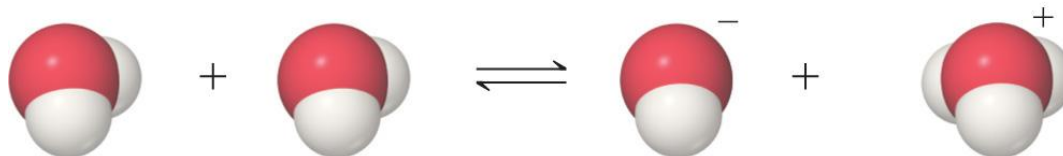
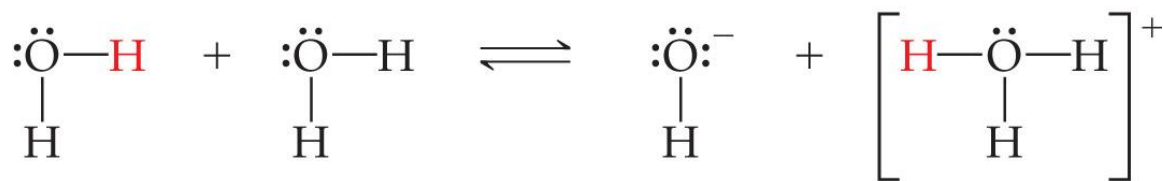
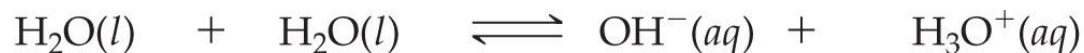
- The substances between the lines with H_2O are conjugate acid–base pairs in water.

Acid and Base Strength

- *In every acid–base reaction, equilibrium favors transfer of the proton from the stronger acid to the stronger base to form the weaker acid and the weaker base.*
- $\text{HCl}(aq) + \text{H}_2\text{O}(l) \rightarrow \text{H}_3\text{O}^+(aq) + \text{Cl}^-(aq)$
- H_2O is a much stronger base than Cl^- , so the equilibrium lies far to the right ($K \gg 1$).
- ❖ $\text{CH}_3\text{COOH}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{CH}_3\text{COO}^-(aq)$
- ❖ Acetate is a stronger base than H_2O , so the equilibrium favors the left side ($K < 1$).

Autoionization of Water

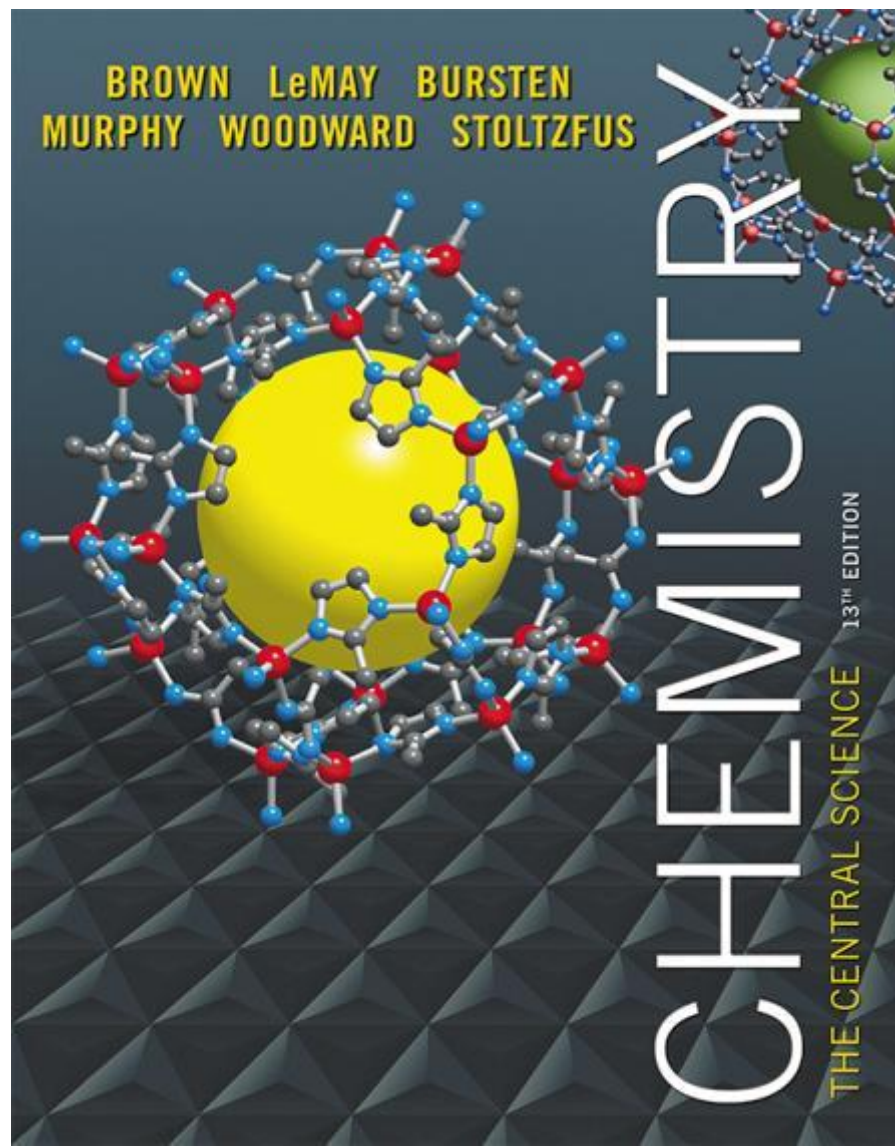
- Water is amphoteric.
- In pure water, a few molecules act as bases and a few act as acids.
- This is referred to as **autoionization**.



Acid

Base

Chapter 4. Acid-Base Equilibria



Reference:

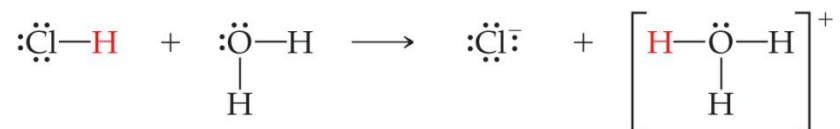
Chapter 16

Acid–Base Equilibria

Brønsted–Lowry Acid and Base

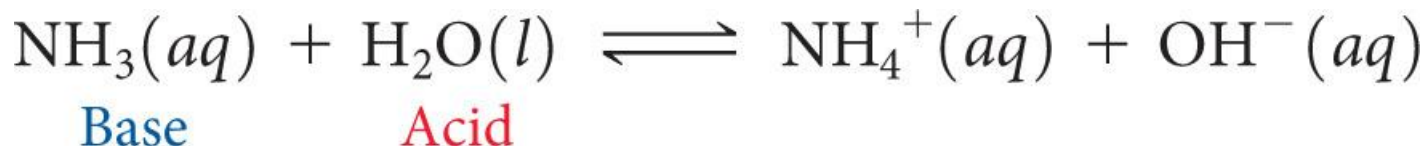
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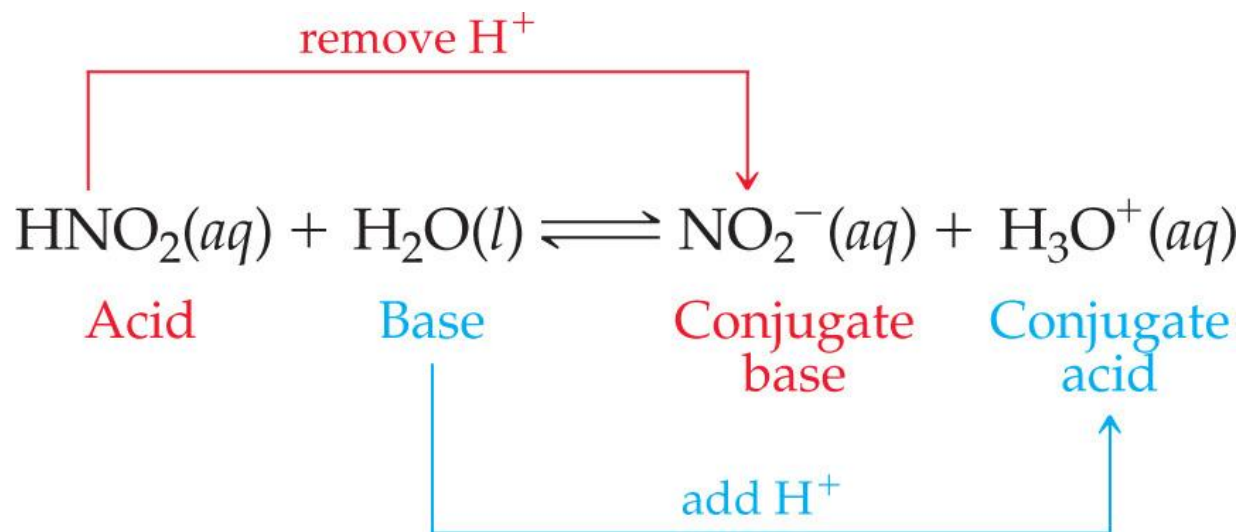
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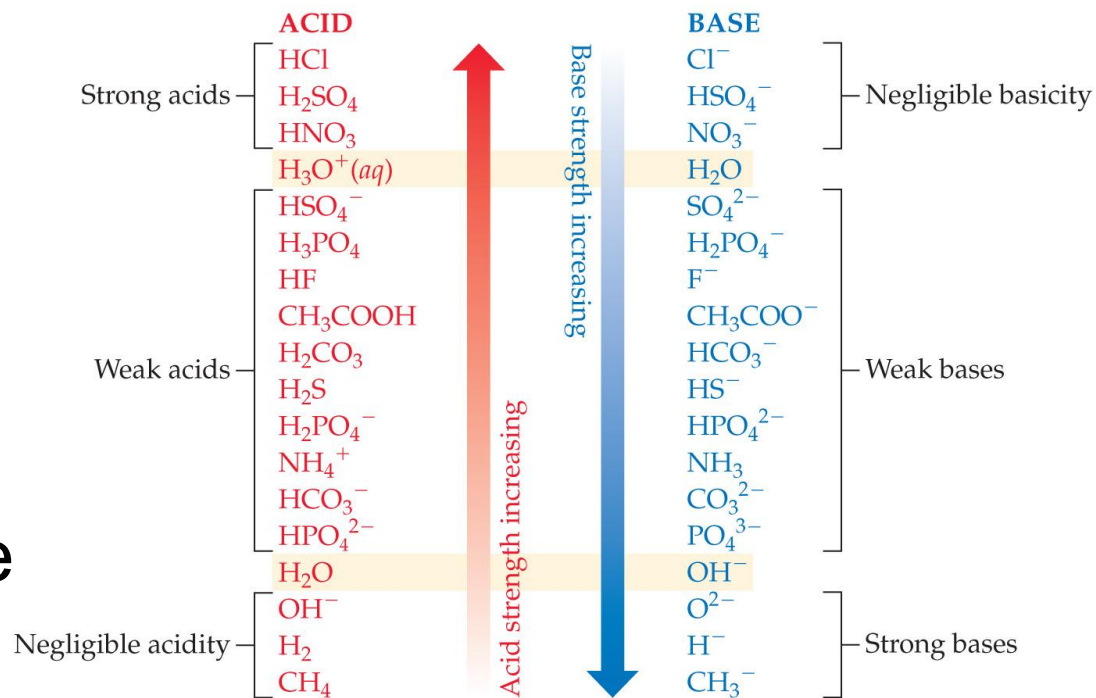
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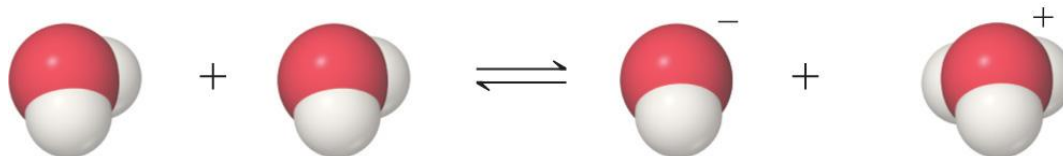
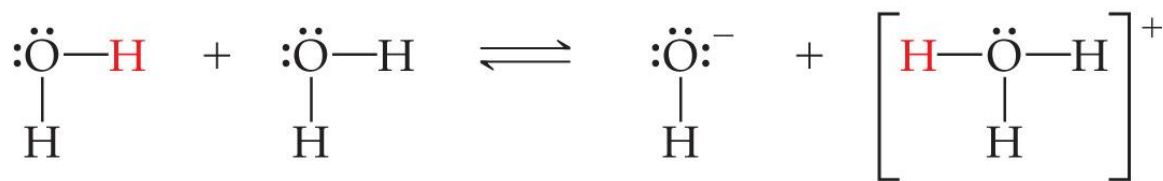
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- H_2O is a much stronger base than Cl^- , so the equilibrium lies far to the right ($K \gg 1$).
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- ❖ Acetate is a stronger base than H_2O , so the equilibrium favors the left side ($K < 1$).

Autoionization of Water

- Water is amphoteric.
- In pure water, a few molecules act as bases and a few act as acids.
- This is referred to as **autoionization**.



Acid

Base

Ion Product Constant

- The equilibrium expression for this process is

$$K_c = [\text{H}_3\text{O}^+][\text{OH}^-]$$

- This special equilibrium constant is referred to as the **ion product constant** for water, K_w .
- At 25 °C, $K_w = 1.0 \times 10^{-14}$

Aqueous Solutions Can Be Acidic, Basic, or Neutral

- If a solution is neutral, $[H^+] = [OH^-]$.
- If a solution is acidic, $[H^+] > [OH^-]$.
- If a solution is basic, $[H^+] < [OH^-]$.



Acidic solution

$$[H^+] > [OH^-]$$

$$[H^+][OH^-] = 1.0 \times 10^{-14}$$



Neutral solution

$$[H^+] = [OH^-]$$

$$[H^+][OH^-] = 1.0 \times 10^{-14}$$



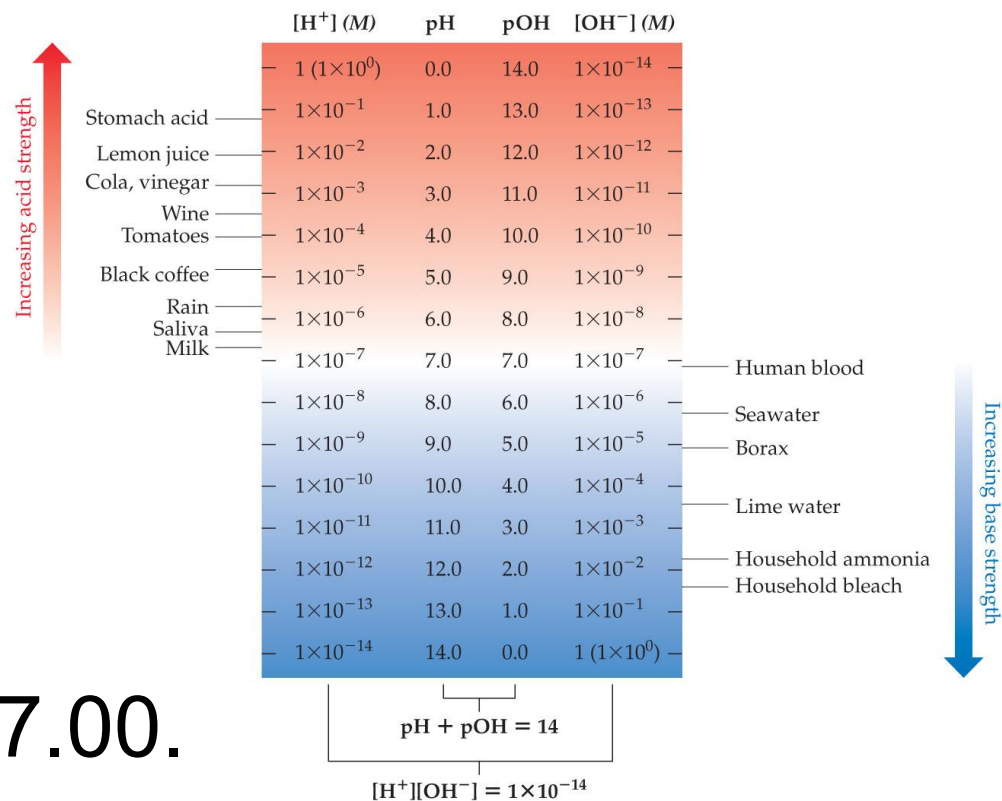
Basic solution

$$[H^+] < [OH^-]$$

$$[H^+][OH^-] = 1.0 \times 10^{-14}$$

pH

- pH is a method of reporting hydrogen ion concentration.
- $\text{pH} = -\log[\text{H}^+]$
- Neutral pH is 7.00.
- Acidic pH is below 7.00.
- Basic pH is above 7.00.



Other “p” Scales

- The “p” in pH tells us to take the $-\log$ of a quantity (in this case, hydrogen ions).
- Some other “p” systems are
 - pOH: $-\log[\text{OH}^-]$
 - $\text{p}K_w$: $-\log K_w$

Relating pH and pOH

Because

$$[\text{H}_3\text{O}^+][\text{OH}^-] = K_w = 1.0 \times 10^{-14}$$

we can take the $-\log$ of the equation

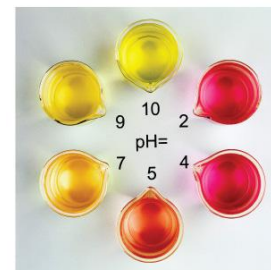
$$-\log[\text{H}_3\text{O}^+] + -\log[\text{OH}^-] = -\log K_w = 14.00$$

which results in

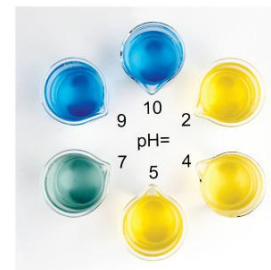
$$\text{pH} + \text{pOH} = \text{p}K_w = 14.00$$

How Do We Measure pH?

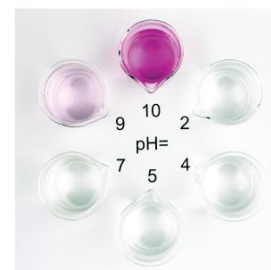
- Indicators, including litmus paper, are used for less accurate measurements; an indicator is one color in its acid form and another color in its basic form.
- pH meters are used for accurate measurement of pH; electrodes indicate small changes in voltage to detect pH.



Methyl red



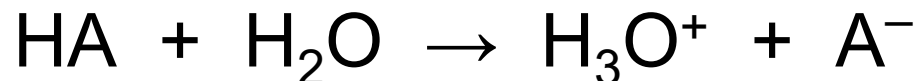
Bromthymol blue



Phenolphthalein

Strong Acids

- You will recall that the seven strong acids are HCl, HBr, HI, HNO₃, H₂SO₄, HClO₃, and HClO₄.
- These are, by definition, strong electrolytes and exist totally as ions in aqueous solution; e.g.,

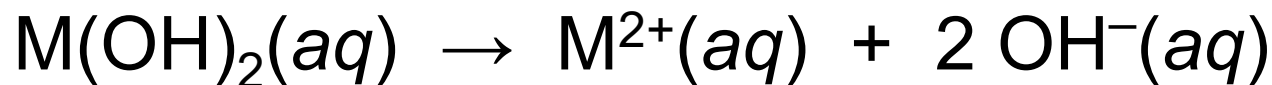


- So, for the monoprotic strong acids,

$$[\text{H}_3\text{O}^+] = [\text{acid}]$$

Strong Bases

- Strong bases are the soluble hydroxides, which are the alkali metal and heavier alkaline earth metal hydroxides (Ca^{2+} , Sr^{2+} , and Ba^{2+}).
- Again, these substances dissociate completely in aqueous solution; e.g.,



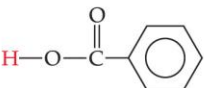
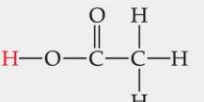
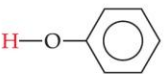
Weak Acids

- For a weak acid, the equation for its dissociation is

$$\text{HA}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{A}^-(aq)$$
- Since it is an equilibrium, there is an equilibrium constant related to it, called the **acid-dissociation constant**, K_a .

$$K_a = [\text{H}_3\text{O}^+][\text{A}^-] / [\text{HA}]$$

Table 16.2 Some Weak Acids in Water at 25 °C

Acid	Structural Formula*	Conjugate Base	K_a
Chlorous (HClO_2)	$\text{H}-\text{O}-\text{Cl}-\text{O}$	ClO_2^-	1.0×10^{-2}
Hydrofluoric (HF)	$\text{H}-\text{F}$	F^-	6.8×10^{-4}
Nitrous (HNO_2)	$\text{H}-\text{O}-\text{N}=\text{O}$	NO_2^-	4.5×10^{-4}
Benzoic ($\text{C}_6\text{H}_5\text{COOH}$)		$\text{C}_6\text{H}_5\text{COO}^-$	6.3×10^{-5}
Acetic (CH_3COOH)		CH_3COO^-	1.8×10^{-5}
Hypochlorous (HOCl)	$\text{H}-\text{O}-\text{Cl}$	OCl^-	3.0×10^{-8}
Hydrocyanic (HCN)	$\text{H}-\text{C}\equiv\text{N}$	CN^-	4.9×10^{-10}
Phenol (HOC_6H_5)		$\text{C}_6\text{H}_5\text{O}^-$	1.3×10^{-10}

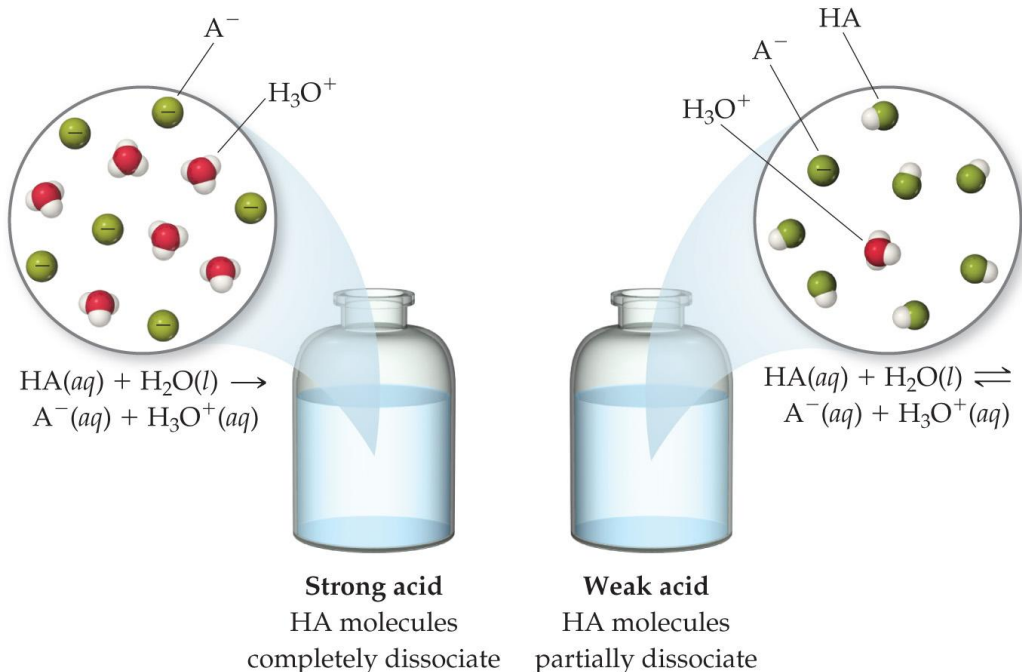
*The proton that ionizes is shown in red.

- The greater the value of K_a , the stronger is the acid.

For HCOOH , $K_a = 1.8 \times 10^{-4}$
 For CF_3COOH , $K_a = 0.56$

Comparing Strong and Weak Acids

- What is present in solution for a strong acid versus a weak acid?
- Strong acids *completely* dissociate to ions.
- Weak acids only *partially* dissociate to ions.





- Strong acids:
- Almost **completely ionized** in solution
- Sulphuric acid, H_2SO_4
- Hydrochloric acid, HCl
- Nitric acid, HNO_3
- From acidic oxides:
- SO_2 , SO_3 , NO_x

- Weak acids:
- **Partially ionized** in solution,
- Hydrofluoric acid, HF
- Acetic acid CH_3COOH
- Formic acid HCOOH
- Benzoic acid $\text{C}_6\text{H}_5\text{COOH}$
- Carbonic acid H_2CO_3
- Phosphoric acid, H_3PO_4
- From: CO_2 , P_2O_3 , P_2O_5 ,

Calculating K_a from the pH

- The pH of a 0.10 *M* solution of formic acid, HCOOH, at 25 °C is 2.38. Calculate K_a for formic acid at this temperature.

➤ We know that

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{HCOO}^-]}{[\text{HCOOH}]}$$

- To calculate K_a , we need the equilibrium concentrations of all three things.
- We can find $[\text{H}_3\text{O}^+]$, which is the same as $[\text{HCOO}^-]$, from the pH.
- $[\text{H}_3\text{O}^+] = [\text{HCOO}^-] = 10^{-2.38} = 4.2 \times 10^{-3}$

Calculating K_a from pH

Now we can set up a table for equilibrium concentrations. We know initial HCOOH (0.10 M) and ion concentrations (0 M); we found equilibrium ion concentrations ($4.2 \times 10^{-3}\text{ M}$); so we calculate the change, then the equilibrium HCOOH concentration.

	[HCOOH], M	[H ₃ O ⁺], M	[HCOO ⁻], M
Initially	0.10	0	0
Change	-4.2×10^{-3}	$+4.2 \times 10^{-3}$	$+4.2 \times 10^{-3}$
At equilibrium	$0.10 - 4.2 \times 10^{-3}$ $= 0.0958 = 0.10$	4.2×10^{-3}	4.2×10^{-3}

Calculating K_a from pH

- This allows us to calculate K_a by putting in the equilibrium concentrations.

$$K_a = \frac{[4.2 \times 10^{-3}][4.2 \times 10^{-3}]}{[0.10]}$$
$$= 1.8 \times 10^{-4}$$

Calculating Percent Ionization

- Percent ionization = $\frac{[\text{H}_3\text{O}^+]_{\text{eq}}}{[\text{HA}]_{\text{initial}}} \times 100$
- In this example,

$$[\text{H}_3\text{O}^+]_{\text{eq}} = 4.2 \times 10^{-3} \text{ M}$$

$$[\text{HCOOH}]_{\text{initial}} = 0.10 \text{ M}$$

$$\begin{aligned}\text{Percent ionization} &= \frac{4.2 \times 10^{-3}}{0.10} \times 100 \\ &= 4.2\%\end{aligned}$$

Method to Follow to Calculate pH Using K_a

- 1) Write the chemical equation for the ionization equilibrium.
- 2) Write the equilibrium constant expression.
- 3) Set up a table for Initial/Change in concentration /Equilibrium Concentration to determine equilibrium concentrations as a function of change (x).
- 4) Substitute equilibrium concentrations into the equilibrium constant expression and solve for x .
(Make assumptions if you can!)

Example

- Calculate the pH of a 0.30 *M* solution of acetic acid, CH₃COOH, at 25 °C.



2) $K_a = [\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$

3)

	CH ₃ COOH (<i>M</i>)	H ₃ O ⁺ (<i>M</i>)	CH ₃ COO ⁻ (<i>M</i>)
Initial Concentration (<i>M</i>)	0.30	0	0
Change in Concentration (<i>M</i>)	- <i>x</i>	+ <i>x</i>	+ <i>x</i>
Equilibrium Concentration (<i>M</i>)	0.30 - <i>x</i>	<i>x</i>	<i>x</i>

Example (concluded)

$$\begin{aligned} 4) \quad K_a &= [\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}] \\ &= (x)(x) / (0.30 - x) \end{aligned}$$

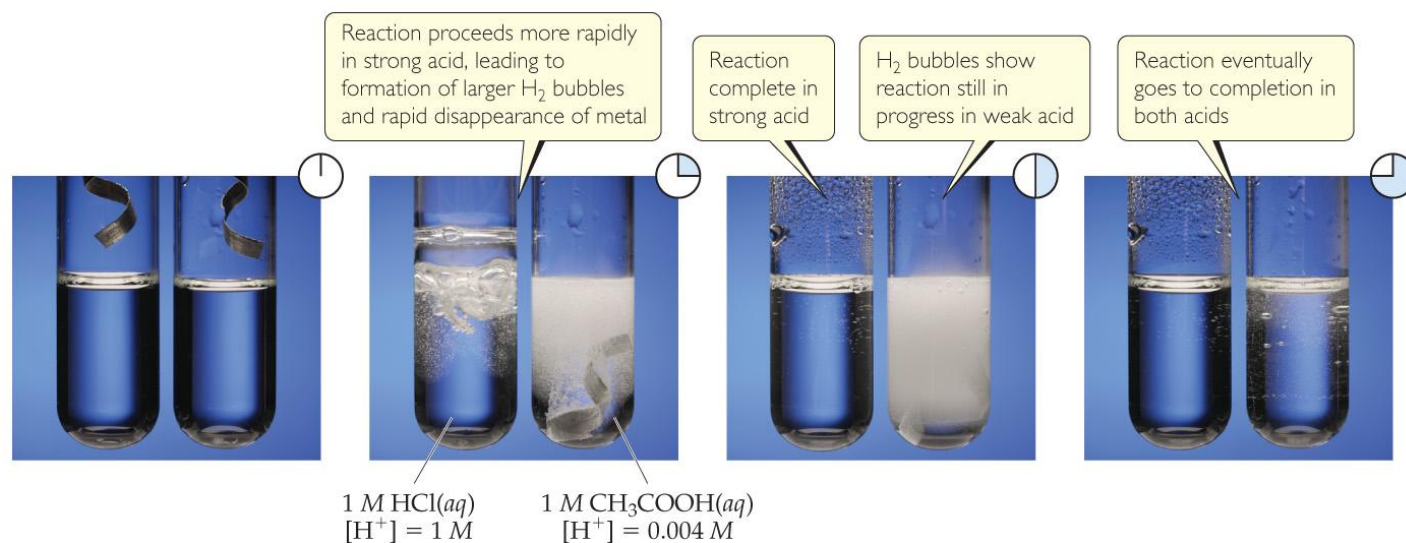
If we assume that $x \ll 0.30$, then $0.30 - x$ becomes 0.30 . The problem becomes easier, since we don't have to use the quadratic formula to solve it.

$$K_a = 1.8 \times 10^{-5} = x^2 / 0.30, \text{ so } x = 2.3 \times 10^{-3}$$

$$x = [\text{H}_3\text{O}^+], \text{ so } \text{pH} = -\log(2.3 \times 10^{-3}) = 2.64$$

Strong vs. Weak Acids— Another Comparison

- Strong Acid: $[\text{H}^+]_{\text{eq}} = [\text{HA}]_{\text{init}}$
- Weak Acid: $[\text{H}^+]_{\text{eq}} < [\text{HA}]_{\text{init}}$
- This creates a difference in conductivity and in rates of chemical reactions.



Polyprotic Acids

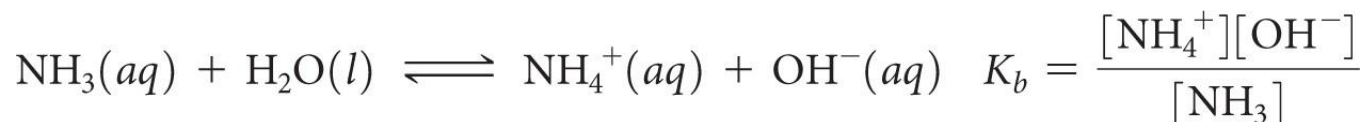
- Polyprotic acids have more than one acidic proton.
- It is always easier to remove the first proton than any successive proton.
- If the factor in the K_a values for the first and second dissociation has a difference of 3 or greater, the pH generally depends *only* on the first dissociation.

Table 16.3 Acid-Dissociation Constants of Some Common Polyprotic Acids

Name	Formula	K_{a1}	K_{a2}	K_{a3}
Ascorbic	$\text{H}_2\text{C}_6\text{H}_6\text{O}_6$	8.0×10^{-5}	1.6×10^{-12}	
Carbonic	H_2CO_3	4.3×10^{-7}	5.6×10^{-11}	
Citric	$\text{H}_3\text{C}_6\text{H}_5\text{O}_7$	7.4×10^{-4}	1.7×10^{-5}	4.0×10^{-7}
Oxalic	$\text{HOOC}-\text{COOH}$	5.9×10^{-2}	6.4×10^{-5}	
Phosphoric	H_3PO_4	7.5×10^{-3}	6.2×10^{-8}	4.2×10^{-13}
Sulfurous	H_2SO_3	1.7×10^{-2}	6.4×10^{-8}	
Sulfuric	H_2SO_4	Large	1.2×10^{-2}	
Tartaric	$\text{C}_2\text{H}_2\text{O}_2(\text{COOH})_2$	1.0×10^{-3}	4.6×10^{-5}	

Weak Bases

- Ammonia, NH_3 , is a weak base.
- Like weak acids, weak bases have an equilibrium constant called the **base dissociation constant**.
- Equilibrium calculations work the *same* as for acids, using the base dissociation constant instead.



Base Dissociation Constants

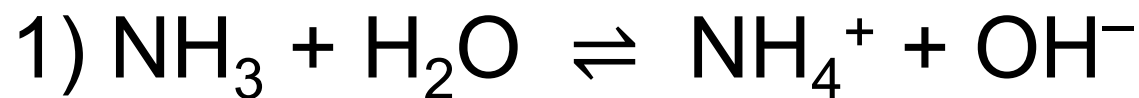
Table 16.4 Some Weak Bases in Water at 25 °C

Base	Structural Formula*	Conjugate Acid	K_b
Ammonia (NH_3)	$\begin{array}{c} \text{H}-\ddot{\text{N}}-\text{H} \\ \\ \text{H} \end{array}$	NH_4^+	1.8×10^{-5}
Pyridine ($\text{C}_5\text{H}_5\text{N}$)		$\text{C}_5\text{H}_5\text{NH}^+$	1.7×10^{-9}
Hydroxylamine (HONH_2)	$\begin{array}{c} \text{H}-\ddot{\text{N}}-\ddot{\text{O}}\text{H} \\ \\ \text{H} \end{array}$	HONH_3^+	1.1×10^{-8}
Methylamine (CH_3NH_2)	$\begin{array}{c} \text{H}-\ddot{\text{N}}-\text{CH}_3 \\ \\ \text{H} \end{array}$	CH_3NH_3^+	4.4×10^{-4}
Hydrosulfide ion (HS^-)	$\left[\text{H}-\ddot{\text{S}}: \right]^-$	H_2S	1.8×10^{-7}
Carbonate ion (CO_3^{2-})	$\left[\begin{array}{c} \ddot{\text{O}}: \\ \\ \text{C} \\ / \quad \backslash \\ \ddot{\text{O}}: \quad \ddot{\text{O}}: \end{array} \right]^{2-}$	HCO_3^-	1.8×10^{-4}
Hypochlorite ion (ClO^-)	$\left[:\ddot{\text{Cl}}-\ddot{\text{O}}: \right]^-$	HClO	3.3×10^{-7}

*The atom that accepts the proton is shown in blue.

Example

- What is the pH of 0.15 M NH_3 ?



2) $K_b = [\text{NH}_4^+][\text{OH}^-] / [\text{NH}_3] = 1.8 \times 10^{-5}$

3)

	$\text{NH}_3 (M)$	$\text{NH}_4^+ (M)$	$\text{OH}^- (M)$
Initial Concentration (M)	0.15	0	0
Change in Concentration (M)	$-x$	$+x$	$+x$
Equilibrium Concentration (M)	$0.15 - x$	x	x

Example (completed)

$$4) 1.8 \times 10^{-5} = x^2 / (0.15 - x)$$

If we assume that $x \ll 0.15$, $0.15 - x = 0.15$.

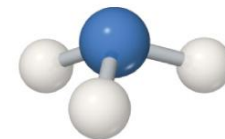
$$\text{Then: } 1.8 \times 10^{-5} = x^2 / 0.15$$

$$\text{and: } x = 1.6 \times 10^{-3}$$

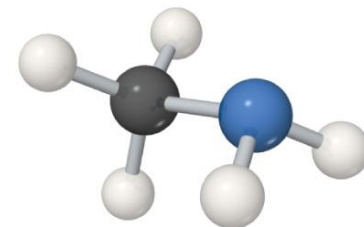
Note: x is the molarity of OH^- , so $-\log(x)$ will be the pOH (pOH = 2.80) and $[14.00 - \text{pOH}]$ is pH (pH = 11.20).

Types of Weak Bases

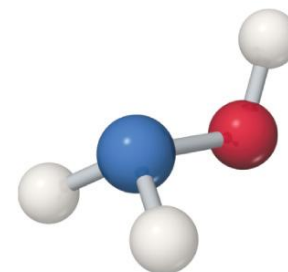
- Two main categories
 - 1) Neutral substances with an atom that has a nonbonding pair of electrons that can accept H^+ (like ammonia and the **amines**)
 - 2) Anions of weak acids



Ammonia
 NH_3



Methylamine
 CH_3NH_2



Hydroxylamine
 NH_2OH

Relationship between K_a and K_b

Table 16.5 Some Conjugate Acid–Base Pairs

Acid	K_a	Base	K_b
HNO ₃	(Strong acid)	NO ₃ [−]	(Negligible basicity)
HF	6.8×10^{-4}	F [−]	1.5×10^{-11}
CH ₃ COOH	1.8×10^{-5}	CH ₃ COO [−]	5.6×10^{-10}
H ₂ CO ₃	4.3×10^{-7}	HCO ₃ [−]	2.3×10^{-8}
NH ₄ ⁺	5.6×10^{-10}	NH ₃	1.8×10^{-5}
HCO ₃ [−]	5.6×10^{-11}	CO ₃ ^{2−}	1.8×10^{-4}
OH [−]	(Negligible acidity)	O ^{2−}	(Strong base)

For a conjugate acid–base pair, K_a and K_b are related in this way:

$$K_a \times K_b = K_w$$

Therefore, if you know one of them, you can calculate the other.

Anions

- Anions of strong acids are neutral. For example, Cl^- will *not* react with water, so OH^- can't be formed.
- Anions of weak acids are conjugate bases, so they create OH^- in water; e.g.,
$$\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$$
- Protonated anions from polyprotic acids can be acids or bases: If $K_a > K_b$, the anion will be acidic; if $K_b > K_a$, the anion will be basic.

Cations

- Group I or Group II (Ca^{2+} , Sr^{2+} , or Ba^{2+}) metal cations are neutral.
- Polyatomic cations are typically the conjugate acids of a weak base; e.g., NH_4^+ .
- Transition and post-transition metal cations are acidic. Why? (There are no H atoms in these cations!)

Hydrated Cations

- Transition and post-transition metals form hydrated cations.
- The water attached to the metal is more acidic than free water molecules, making the hydrated ions acidic.

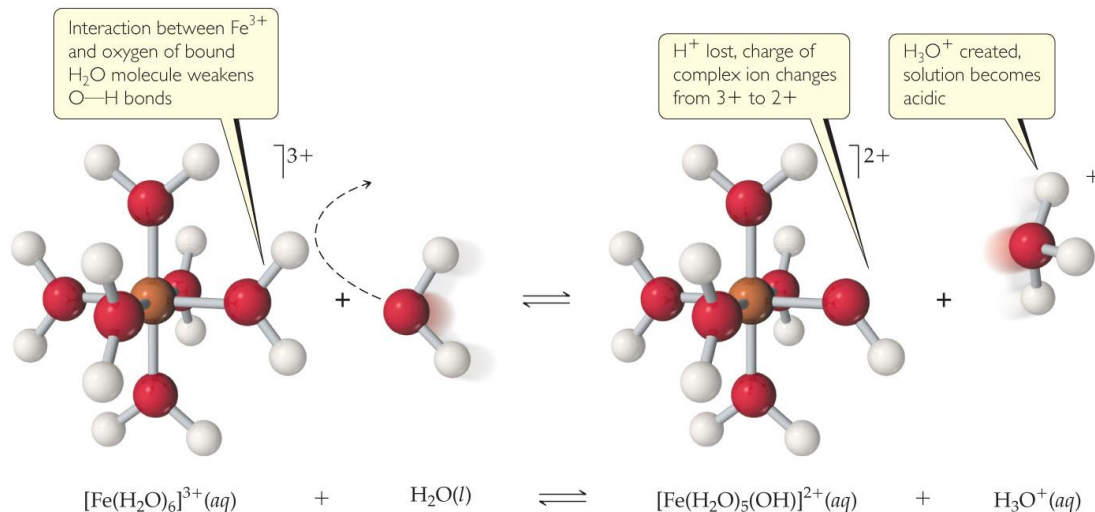


Table 16.6 Acid-Dissociation Constants for Metal Cations in Aqueous Solution at 25 °C

Cation	K_a
Fe^{2+}	3.2×10^{-10}
Zn^{2+}	2.5×10^{-10}
Ni^{2+}	2.5×10^{-11}
Fe^{3+}	6.3×10^{-3}
Cr^{3+}	1.6×10^{-4}
Al^{3+}	1.4×10^{-5}

Factors that Affect Acid Strength

- 1) H—A bond must be polarized with $\delta+$ on the H atom and $\delta-$ on the A atom
- 2) Bond strength: Weaker bonds can be broken more easily, making the acid stronger.
- 3) Stability of A^- : More stable anion means stronger acid.

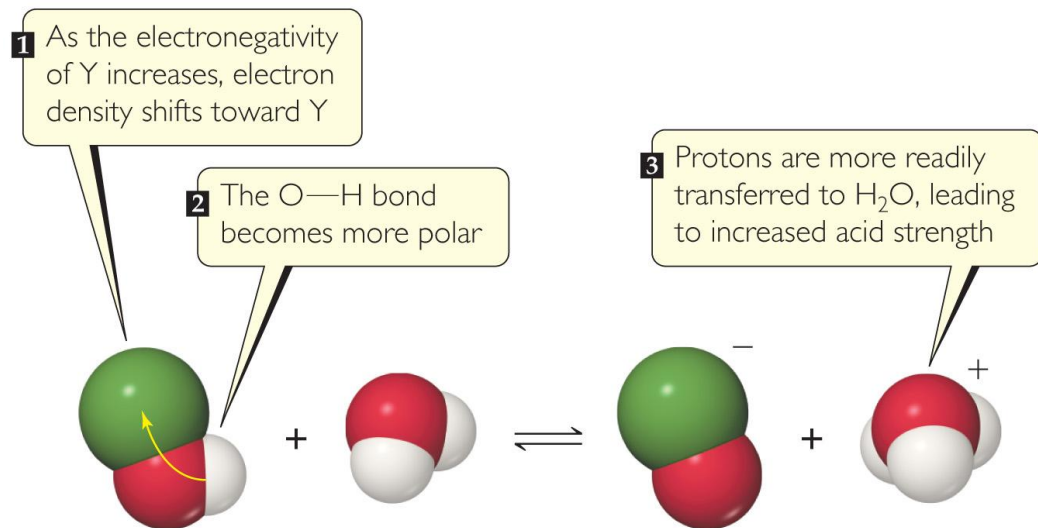
Binary Acids

- Binary acids consist of H and one other element.
- Within a group, H—A **bond strength** is generally the most important factor.
- Within a period, **bond polarity** is the most important factor to determine acid strength.

4A	5A	6A	7A	
CH₄ Neither acid nor base	NH₃ Weak base $K_b = 1.8 \times 10^{-5}$	H₂O	HF Weak acid $K_a = 6.8 \times 10^{-4}$	Increasing acid strength ↓
SiH₄ Neither acid nor base	PH₃ Very weak base $K_b = 4 \times 10^{-28}$	H₂S Weak acid $K_a = 9.5 \times 10^{-8}$	HCl Strong acid	
		H₂Se Weak acid $K_a = 1.3 \times 10^{-4}$	HBr Strong acid	
Increasing acid strength →				

Oxyacids

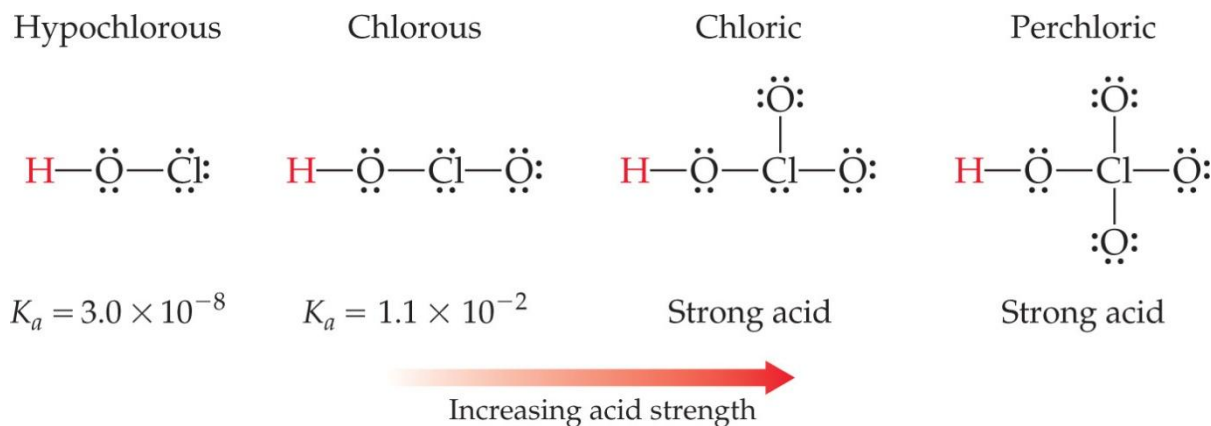
- **Oxyacids** consist of H, O, and one other element, which is a nonmetal.
- Generally, as the electronegativity of the nonmetal increases, the acidity increases for acids with the same structure.



Substance	Y—OH	Electronegativity of Y	Dissociation constant
Hypochlorous acid	Cl—OH	3.0	$K_a = 3.0 \times 10^{-8}$
Hypobromous acid	Br—OH	2.8	$K_a = 2.5 \times 10^{-9}$
Hypoiodous acid	I—OH	2.5	$K_a = 2.3 \times 10^{-11}$
Water	H—OH	2.1	$K_w = 1.0 \times 10^{-14}$

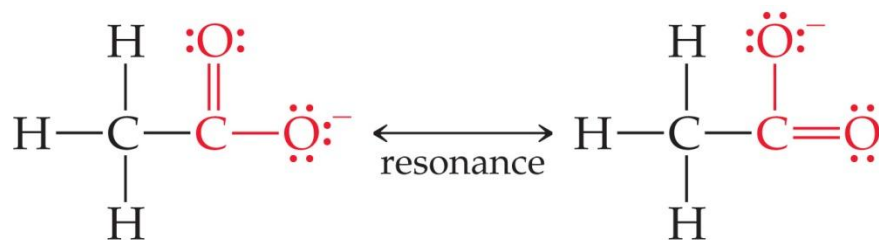
Oxyacids with Same “Other” Element

- If an element can form more than one oxyacid, the oxyacid with more O atoms is more acidic; e.g., sulfuric acid versus sulfurous acid.
- Another way of saying it: If the oxidation number increases, the acidity increases.



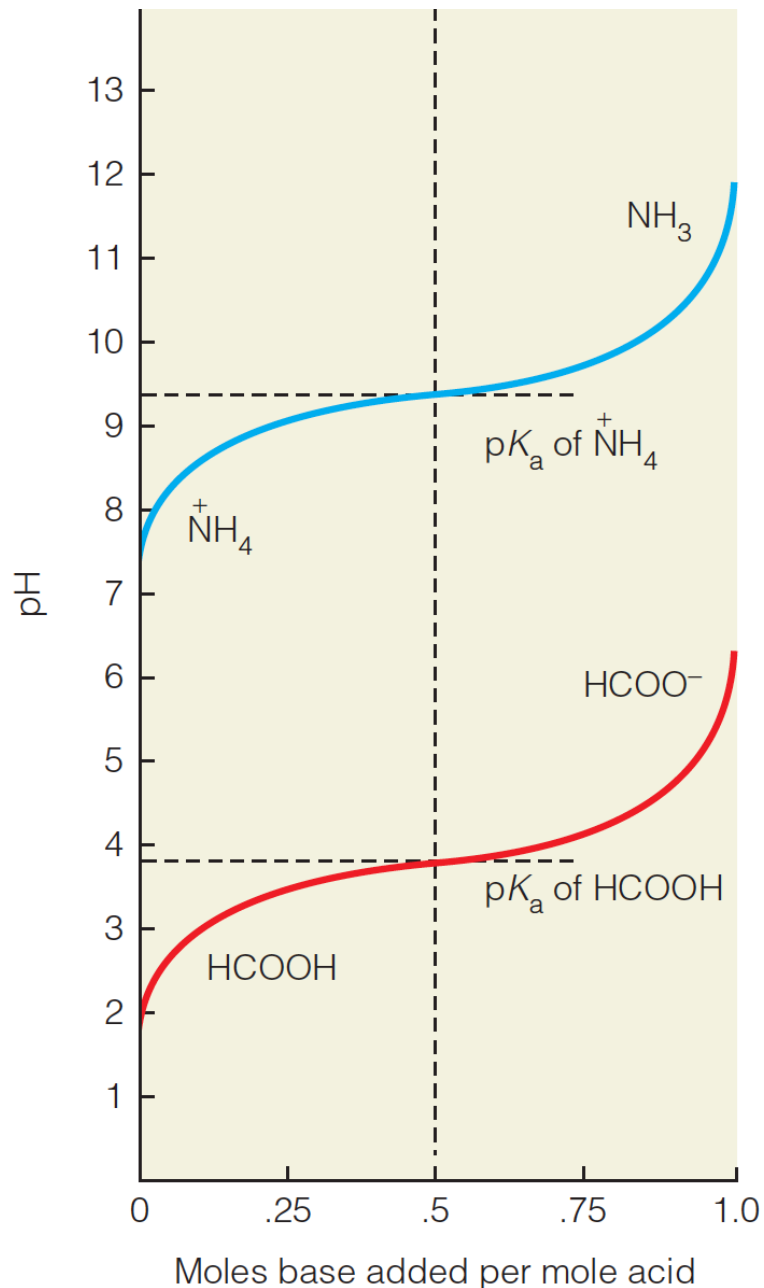
Carboxylic Acids

- **Carboxylic acids** are organic acids containing the —COOH group.
- Factors contributing to their acidic behavior:
 - Other O attached to C draws electron density from O—H bond, increasing polarity.
 - Its conjugate base (carboxylate anion) has resonance forms to stabilize the anion.



The **Henderson–Hasselbalch equation** describes the change in pH during titration of a weak acid or a weak base.

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$



Ionic Equilibria

Buffered solutions are able to minimize the change in pH following addition of acid or base because the conjugate acid (HA) and conjugate base (A^-) of the buffering compound **are present in sufficient** concentration to combine with the added H^+ or OH^- and neutralize them.

Buffer solutions function because the pH of a weak acid–base solution is least sensitive to added acid or base **near the pK_a** , *where the* conjugate acid and conjugate base of the buffer are both present in nearly ***equimolar concentrations***.

Ionic Equilibria

- Organisms must maintain the pH inside cells and in most bodily fluids within the narrow pH range of about 6.5 to 8.0.
- The dihydrogen phosphate–hydrogen phosphate system, with a pK_a of 6.86, plays a major role in controlling *intracellular* pH because phosphate is abundant in cells.
- **Blood** contains dissolved CO_2 as a waste product of metabolism, therefore the carbonic acid–bicarbonate system provides considerable buffering capacity.

Ion Product Constant

- The equilibrium expression for this process is

$$K_c = [\text{H}_3\text{O}^+][\text{OH}^-]$$

- This special equilibrium constant is referred to as the **ion product constant** for water, K_w .
- At 25 °C, $K_w = 1.0 \times 10^{-14}$

Aqueous Solutions Can Be Acidic, Basic, or Neutral

- If a solution is neutral, $[\text{H}^+] = [\text{OH}^-]$.
- If a solution is acidic, $[\text{H}^+] > [\text{OH}^-]$.
- If a solution is basic, $[\text{H}^+] < [\text{OH}^-]$.



Acidic solution

$$[\text{H}^+] > [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$



Neutral solution

$$[\text{H}^+] = [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$



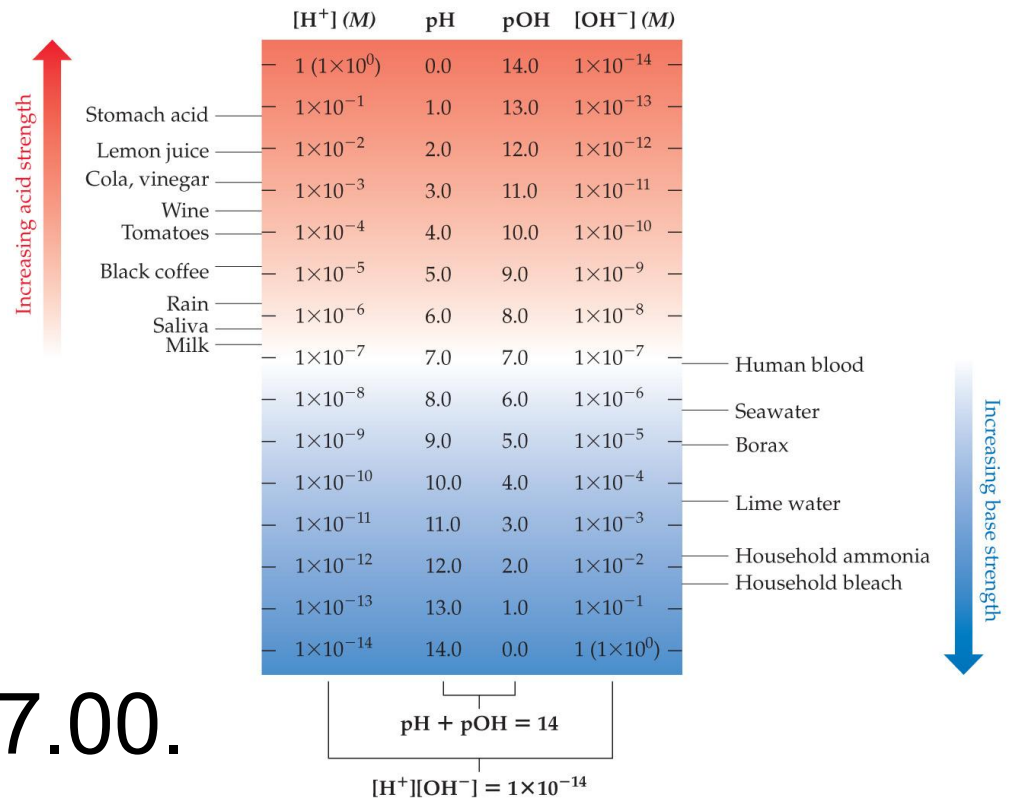
Basic solution

$$[\text{H}^+] < [\text{OH}^-]$$

$$[\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

pH

- pH is a method of reporting hydrogen ion concentration.
- $\text{pH} = -\log[\text{H}^+]$
- Neutral pH is 7.00.
- Acidic pH is below 7.00.
- Basic pH is above 7.00.



Other “p” Scales

- The “p” in pH tells us to take the $-\log$ of a quantity (in this case, hydrogen ions).
- Some other “p” systems are
 - pOH: $-\log[\text{OH}^-]$
 - $\text{p}K_w$: $-\log K_w$

Relating pH and pOH

Because

$$[\text{H}_3\text{O}^+][\text{OH}^-] = K_w = 1.0 \times 10^{-14}$$

we can take the $-\log$ of the equation

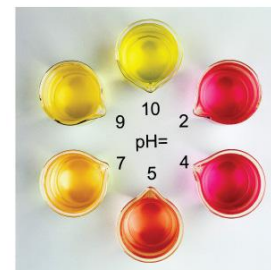
$$-\log[\text{H}_3\text{O}^+] + -\log[\text{OH}^-] = -\log K_w = 14.00$$

which results in

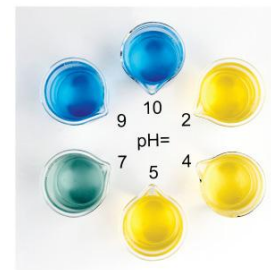
$$\text{pH} + \text{pOH} = \text{p}K_w = 14.00$$

How Do We Measure pH?

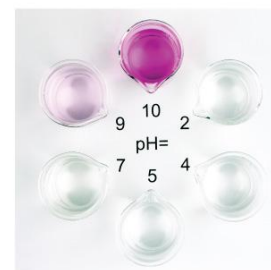
- Indicators, including litmus paper, are used for less accurate measurements; an indicator is one color in its acid form and another color in its basic form.
- pH meters are used for accurate measurement of pH; electrodes indicate small changes in voltage to detect pH.



Methyl red



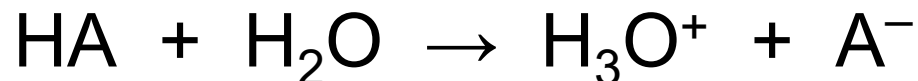
Bromthymol blue



Phenolphthalein

Strong Acids

- You will recall that the seven strong acids are HCl, HBr, HI, HNO₃, H₂SO₄, HClO₃, and HClO₄.
- These are, by definition, strong electrolytes and exist totally as ions in aqueous solution; e.g.,

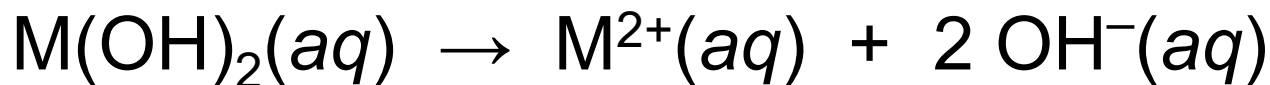


- So, for the monoprotic strong acids,

$$[\text{H}_3\text{O}^+] = [\text{acid}]$$

Strong Bases

- Strong bases are the soluble hydroxides, which are the alkali metal and heavier alkaline earth metal hydroxides (Ca^{2+} , Sr^{2+} , and Ba^{2+}).
- Again, these substances dissociate completely in aqueous solution; e.g.,



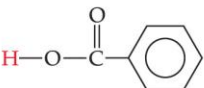
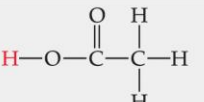
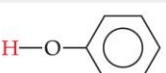
Weak Acids

- For a weak acid, the equation for its dissociation is

$$\text{HA}(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{A}^-(aq)$$
- Since it is an equilibrium, there is an equilibrium constant related to it, called the **acid-dissociation constant, K_a** .

$$K_a = [\text{H}_3\text{O}^+][\text{A}^-] / [\text{HA}]$$

Table 16.2 Some Weak Acids in Water at 25 °C

Acid	Structural Formula*	Conjugate Base	K_a
Chlorous (HClO_2)	$\text{H}-\text{O}-\text{Cl}-\text{O}$	ClO_2^-	1.0×10^{-2}
Hydrofluoric (HF)	$\text{H}-\text{F}$	F^-	6.8×10^{-4}
Nitrous (HNO_2)	$\text{H}-\text{O}-\text{N}=\text{O}$	NO_2^-	4.5×10^{-4}
Benzoic ($\text{C}_6\text{H}_5\text{COOH}$)		$\text{C}_6\text{H}_5\text{COO}^-$	6.3×10^{-5}
Acetic (CH_3COOH)		CH_3COO^-	1.8×10^{-5}
Hypochlorous (HOCl)	$\text{H}-\text{O}-\text{Cl}$	OCl^-	3.0×10^{-8}
Hydrocyanic (HCN)	$\text{H}-\text{C}\equiv\text{N}$	CN^-	4.9×10^{-10}
Phenol (HOC_6H_5)		$\text{C}_6\text{H}_5\text{O}^-$	1.3×10^{-10}

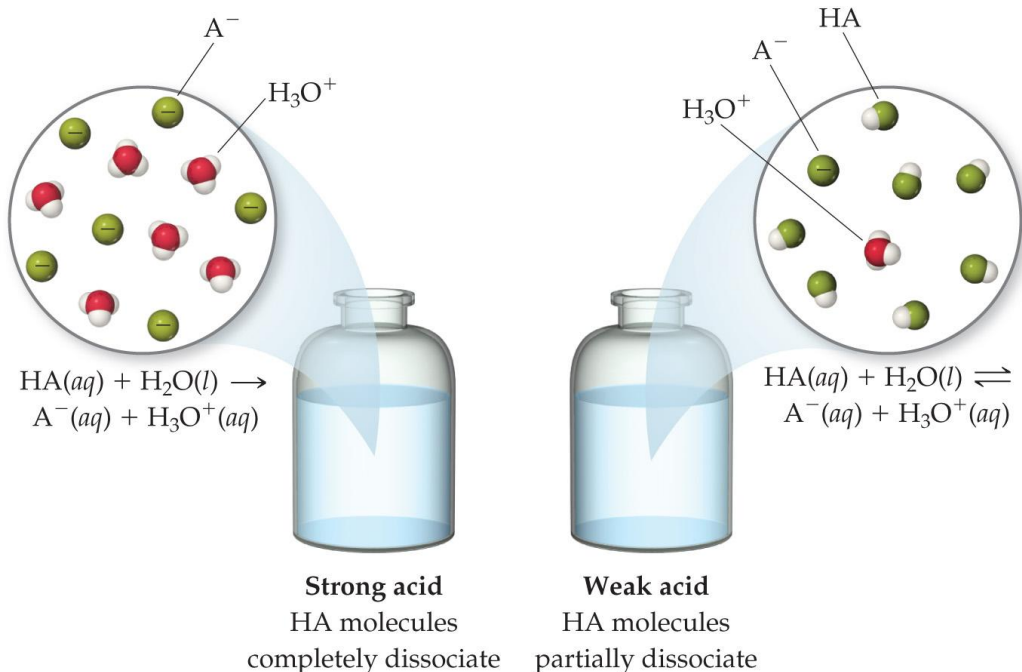
*The proton that ionizes is shown in red.

- The greater the value of K_a , the stronger is the acid.

For HCOOH , $K_a = 1.8 \times 10^{-4}$
 For CF_3COOH , $K_a = 0.56$

Comparing Strong and Weak Acids

- What is present in solution for a strong acid versus a weak acid?
- Strong acids *completely* dissociate to ions.
- Weak acids only *partially* dissociate to ions.





- Strong acids:
- Almost **completely ionized** in solution
- Sulphuric acid, H_2SO_4
- Hydrochloric acid, HCl
- Nitric acid, HNO_3
- From acidic oxides:
- SO_2 , SO_3 , NO_x

- Weak acids:
- **Partially ionized** in solution,
- Hydrofluoric acid, HF
- Acetic acid CH_3COOH
- Formic acid HCOOH
- Benzoic acid $\text{C}_6\text{H}_5\text{COOH}$
- Carbonic acid H_2CO_3
- Phosphoric acid, H_3PO_4
- From: CO_2 , P_2O_3 , P_2O_5 ,

Calculating K_a from the pH

- The pH of a 0.10 *M* solution of formic acid, HCOOH, at 25 °C is 2.38. Calculate K_a for formic acid at this temperature.

➤ We know that

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{HCOO}^-]}{[\text{HCOOH}]}$$

- To calculate K_a , we need the equilibrium concentrations of all three things.
- We can find $[\text{H}_3\text{O}^+]$, which is the same as $[\text{HCOO}^-]$, from the pH.
- $[\text{H}_3\text{O}^+] = [\text{HCOO}^-] = 10^{-2.38} = 4.2 \times 10^{-3}$

Calculating K_a from pH

Now we can set up a table for equilibrium concentrations. We know initial HCOOH (0.10 M) and ion concentrations (0 M); we found equilibrium ion concentrations ($4.2 \times 10^{-3}\text{ M}$); so we calculate the change, then the equilibrium HCOOH concentration.

	[HCOOH], M	[H ₃ O ⁺], M	[HCOO ⁻], M
Initially	0.10	0	0
Change	-4.2×10^{-3}	$+4.2 \times 10^{-3}$	$+4.2 \times 10^{-3}$
At equilibrium	$0.10 - 4.2 \times 10^{-3}$ $= 0.0958 = 0.10$	4.2×10^{-3}	4.2×10^{-3}

Calculating K_a from pH

- This allows us to calculate K_a by putting in the equilibrium concentrations.

$$K_a = \frac{[4.2 \times 10^{-3}][4.2 \times 10^{-3}]}{[0.10]}$$
$$= 1.8 \times 10^{-4}$$

Calculating Percent Ionization

- Percent ionization = $\frac{[\text{H}_3\text{O}^+]_{\text{eq}}}{[\text{HA}]_{\text{initial}}} \times 100$
- In this example,

$$[\text{H}_3\text{O}^+]_{\text{eq}} = 4.2 \times 10^{-3} \text{ M}$$

$$[\text{HCOOH}]_{\text{initial}} = 0.10 \text{ M}$$

$$\begin{aligned}\text{Percent ionization} &= \frac{4.2 \times 10^{-3}}{0.10} \times 100 \\ &= 4.2\%\end{aligned}$$

Method to Follow to Calculate pH Using K_a

- 1) Write the chemical equation for the ionization equilibrium.
- 2) Write the equilibrium constant expression.
- 3) Set up a table for Initial/Change in concentration /Equilibrium Concentration to determine equilibrium concentrations as a function of change (x).
- 4) Substitute equilibrium concentrations into the equilibrium constant expression and solve for x .
(Make assumptions if you can!)

Example

- Calculate the pH of a 0.30 *M* solution of acetic acid, CH₃COOH, at 25 °C.



2) $K_a = [\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}]$

3)

	CH ₃ COOH (<i>M</i>)	H ₃ O ⁺ (<i>M</i>)	CH ₃ COO ⁻ (<i>M</i>)
Initial Concentration (<i>M</i>)	0.30	0	0
Change in Concentration (<i>M</i>)	- <i>x</i>	+ <i>x</i>	+ <i>x</i>
Equilibrium Concentration (<i>M</i>)	0.30 - <i>x</i>	<i>x</i>	<i>x</i>

Example (concluded)

$$\begin{aligned} 4) \quad K_a &= [\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-] / [\text{CH}_3\text{COOH}] \\ &= (x)(x) / (0.30 - x) \end{aligned}$$

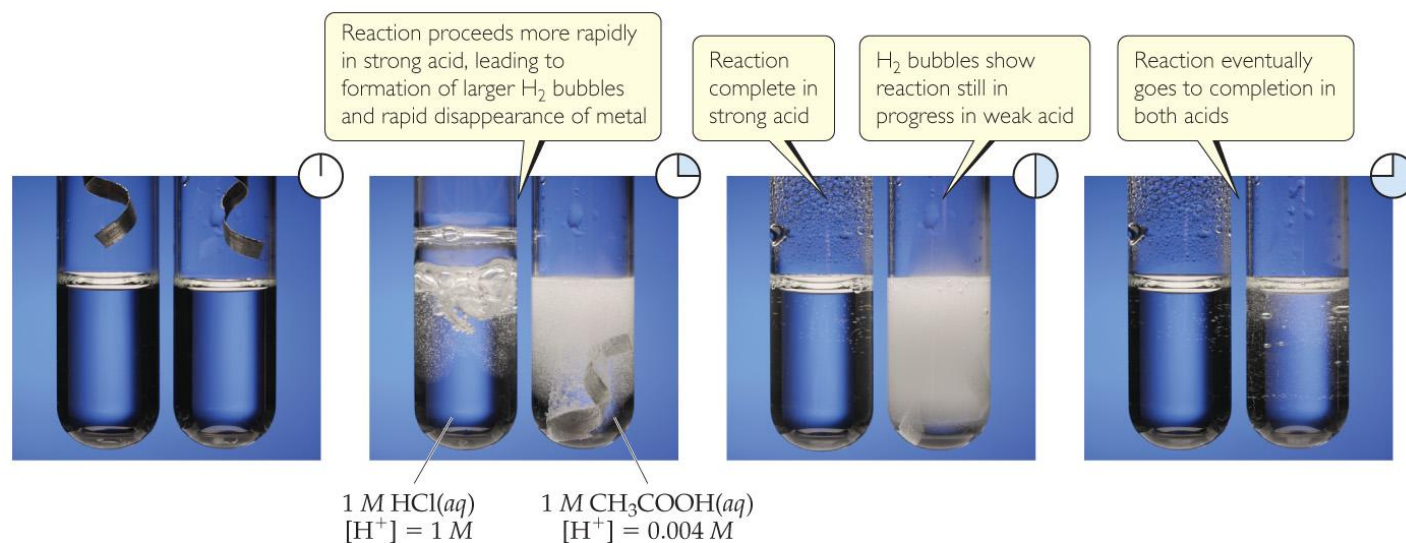
If we assume that $x \ll 0.30$, then $0.30 - x$ becomes 0.30 . The problem becomes easier, since we don't have to use the quadratic formula to solve it.

$$K_a = 1.8 \times 10^{-5} = x^2 / 0.30, \text{ so } x = 2.3 \times 10^{-3}$$

$$x = [\text{H}_3\text{O}^+], \text{ so } \text{pH} = -\log(2.3 \times 10^{-3}) = 2.64$$

Strong vs. Weak Acids— Another Comparison

- Strong Acid: $[\text{H}^+]_{\text{eq}} = [\text{HA}]_{\text{init}}$
- Weak Acid: $[\text{H}^+]_{\text{eq}} < [\text{HA}]_{\text{init}}$
- This creates a difference in conductivity and in rates of chemical reactions.



Polyprotic Acids

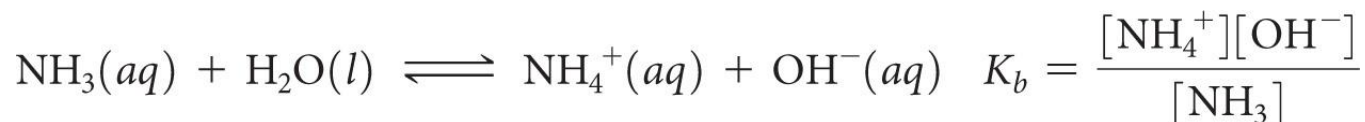
- Polyprotic acids have more than one acidic proton.
- It is always easier to remove the first proton than any successive proton.
- If the factor in the K_a values for the first and second dissociation has a difference of 3 or greater, the pH generally depends *only* on the first dissociation.

Table 16.3 Acid-Dissociation Constants of Some Common Polyprotic Acids

Name	Formula	K_{a1}	K_{a2}	K_{a3}
Ascorbic	$\text{H}_2\text{C}_6\text{H}_6\text{O}_6$	8.0×10^{-5}	1.6×10^{-12}	
Carbonic	H_2CO_3	4.3×10^{-7}	5.6×10^{-11}	
Citric	$\text{H}_3\text{C}_6\text{H}_5\text{O}_7$	7.4×10^{-4}	1.7×10^{-5}	4.0×10^{-7}
Oxalic	$\text{HOOC}-\text{COOH}$	5.9×10^{-2}	6.4×10^{-5}	
Phosphoric	H_3PO_4	7.5×10^{-3}	6.2×10^{-8}	4.2×10^{-13}
Sulfurous	H_2SO_3	1.7×10^{-2}	6.4×10^{-8}	
Sulfuric	H_2SO_4	Large	1.2×10^{-2}	
Tartaric	$\text{C}_2\text{H}_2\text{O}_2(\text{COOH})_2$	1.0×10^{-3}	4.6×10^{-5}	

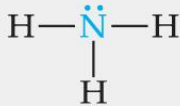
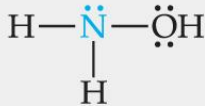
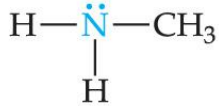
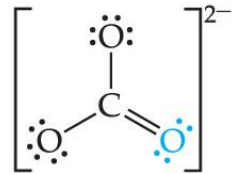
Weak Bases

- Ammonia, NH_3 , is a weak base.
- Like weak acids, weak bases have an equilibrium constant called the **base dissociation constant**.
- Equilibrium calculations work the *same* as for acids, using the base dissociation constant instead.



Base Dissociation Constants

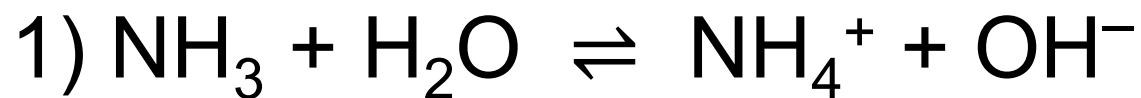
Table 16.4 Some Weak Bases in Water at 25 °C

Base	Structural Formula*	Conjugate Acid	K_b
Ammonia (NH_3)		NH_4^+	1.8×10^{-5}
Pyridine ($\text{C}_5\text{H}_5\text{N}$)		$\text{C}_5\text{H}_5\text{NH}^+$	1.7×10^{-9}
Hydroxylamine (HONH_2)		HONH_3^+	1.1×10^{-8}
Methylamine (CH_3NH_2)		CH_3NH_3^+	4.4×10^{-4}
Hydrosulfide ion (HS^-)		H_2S	1.8×10^{-7}
Carbonate ion (CO_3^{2-})		HCO_3^-	1.8×10^{-4}
Hypochlorite ion (ClO^-)		HClO	3.3×10^{-7}

*The atom that accepts the proton is shown in blue.

Example

- What is the pH of 0.15 M NH_3 ?



2) $K_b = [\text{NH}_4^+][\text{OH}^-] / [\text{NH}_3] = 1.8 \times 10^{-5}$

3)

	$\text{NH}_3 (M)$	$\text{NH}_4^+ (M)$	$\text{OH}^- (M)$
Initial Concentration (M)	0.15	0	0
Change in Concentration (M)	$-x$	$+x$	$+x$
Equilibrium Concentration (M)	$0.15 - x$	x	x

Example (completed)

$$4) 1.8 \times 10^{-5} = x^2 / (0.15 - x)$$

If we assume that $x \ll 0.15$, $0.15 - x = 0.15$.

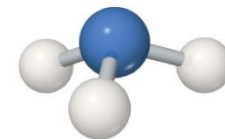
$$\text{Then: } 1.8 \times 10^{-5} = x^2 / 0.15$$

$$\text{and: } x = 1.6 \times 10^{-3}$$

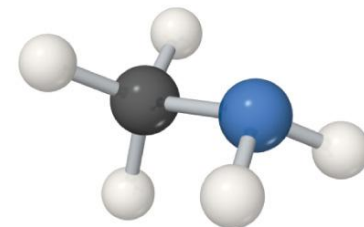
Note: x is the molarity of OH^- , so $-\log(x)$ will be the pOH (pOH = 2.80) and $[14.00 - \text{pOH}]$ is pH (pH = 11.20).

Types of Weak Bases

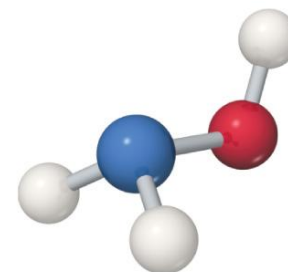
- Two main categories
 - 1) Neutral substances with an atom that has a nonbonding pair of electrons that can accept H^+ (like ammonia and the **amines**)
 - 2) Anions of weak acids



Ammonia
 NH_3



Methylamine
 CH_3NH_2



Hydroxylamine
 NH_2OH

Relationship between K_a and K_b

Table 16.5 Some Conjugate Acid–Base Pairs

Acid	K_a	Base	K_b
HNO ₃	(Strong acid)	NO ₃ [−]	(Negligible basicity)
HF	6.8×10^{-4}	F [−]	1.5×10^{-11}
CH ₃ COOH	1.8×10^{-5}	CH ₃ COO [−]	5.6×10^{-10}
H ₂ CO ₃	4.3×10^{-7}	HCO ₃ [−]	2.3×10^{-8}
NH ₄ ⁺	5.6×10^{-10}	NH ₃	1.8×10^{-5}
HCO ₃ [−]	5.6×10^{-11}	CO ₃ ^{2−}	1.8×10^{-4}
OH [−]	(Negligible acidity)	O ^{2−}	(Strong base)

For a conjugate acid–base pair, K_a and K_b are related in this way:

$$K_a \times K_b = K_w$$

Therefore, if you know one of them, you can calculate the other.

Anions

- Anions of strong acids are neutral. For example, Cl^- will *not* react with water, so OH^- can't be formed.
- Anions of weak acids are conjugate bases, so they create OH^- in water; e.g.,
$$\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$$
- Protonated anions from polyprotic acids can be acids or bases: If $K_a > K_b$, the anion will be acidic; if $K_b > K_a$, the anion will be basic.

Cations

- Group I or Group II (Ca^{2+} , Sr^{2+} , or Ba^{2+}) metal cations are neutral.
- Polyatomic cations are typically the conjugate acids of a weak base; e.g., NH_4^+ .
- Transition and post-transition metal cations are acidic. Why? (There are no H atoms in these cations!)

Hydrated Cations

- Transition and post-transition metals form hydrated cations.
- The water attached to the metal is more acidic than free water molecules, making the hydrated ions acidic.

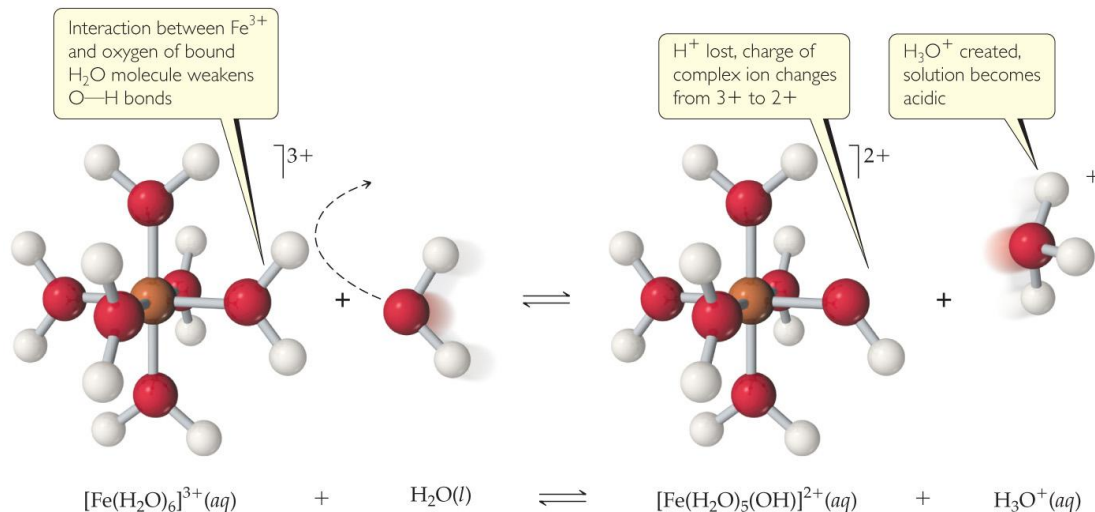


Table 16.6 Acid-Dissociation Constants for Metal Cations in Aqueous Solution at 25 °C

Cation	K_a
Fe^{2+}	3.2×10^{-10}
Zn^{2+}	2.5×10^{-10}
Ni^{2+}	2.5×10^{-11}
Fe^{3+}	6.3×10^{-3}
Cr^{3+}	1.6×10^{-4}
Al^{3+}	1.4×10^{-5}

Factors that Affect Acid Strength

- 1) H—A bond must be polarized with $\delta+$ on the H atom and $\delta-$ on the A atom
- 2) Bond strength: Weaker bonds can be broken more easily, making the acid stronger.
- 3) Stability of A^- : More stable anion means stronger acid.

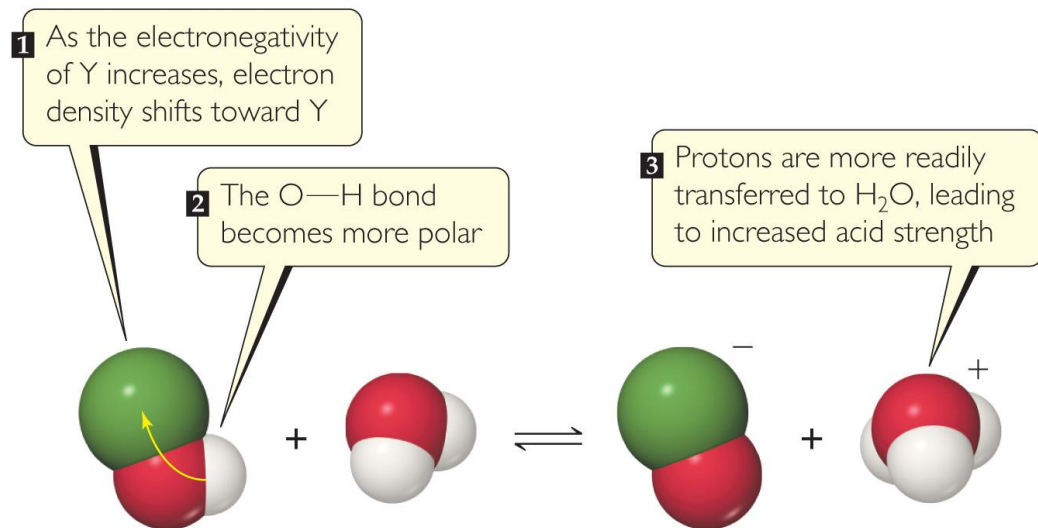
Binary Acids

- Binary acids consist of H and one other element.
- Within a group, H—A **bond strength** is generally the most important factor.
- Within a period, **bond polarity** is the most important factor to determine acid strength.

4A	5A	6A	7A	
CH₄ Neither acid nor base	NH₃ Weak base $K_b = 1.8 \times 10^{-5}$	H₂O	HF Weak acid $K_a = 6.8 \times 10^{-4}$	Increasing acid strength ↓
SiH₄ Neither acid nor base	PH₃ Very weak base $K_b = 4 \times 10^{-28}$	H₂S Weak acid $K_a = 9.5 \times 10^{-8}$	HCl Strong acid	
		H₂Se Weak acid $K_a = 1.3 \times 10^{-4}$	HBr Strong acid	
Increasing acid strength →				

Oxyacids

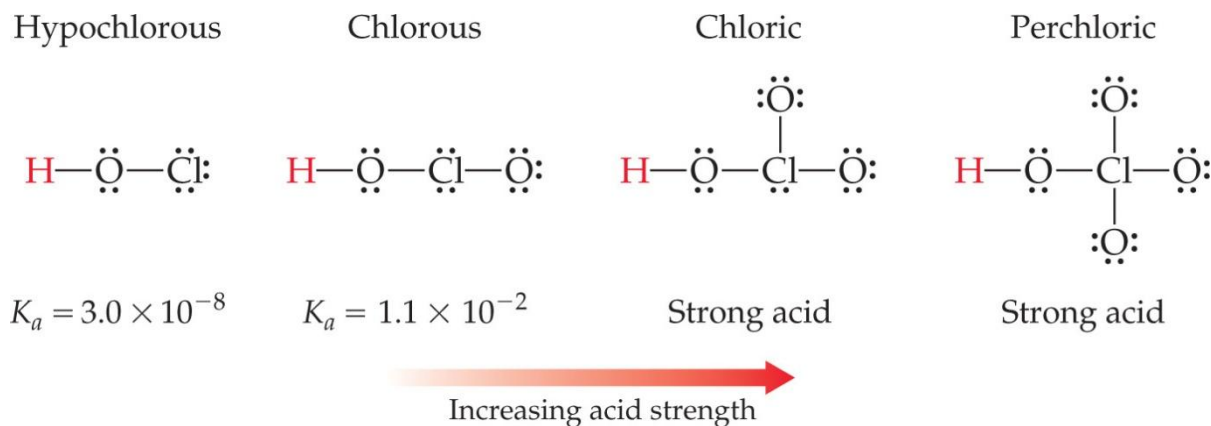
- **Oxyacids** consist of H, O, and one other element, which is a nonmetal.
- Generally, as the electronegativity of the nonmetal increases, the acidity increases for acids with the same structure.



Substance	Y—OH	Electronegativity of Y	Dissociation constant
Hypochlorous acid	Cl—OH	3.0	$K_a = 3.0 \times 10^{-8}$
Hypobromous acid	Br—OH	2.8	$K_a = 2.5 \times 10^{-9}$
Hypoiodous acid	I—OH	2.5	$K_a = 2.3 \times 10^{-11}$
Water	H—OH	2.1	$K_w = 1.0 \times 10^{-14}$

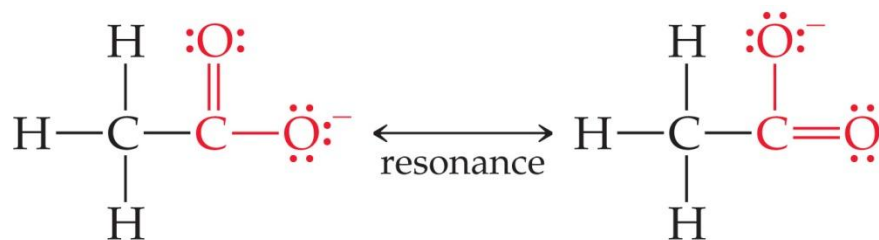
Oxyacids with Same “Other” Element

- If an element can form more than one oxyacid, the oxyacid with more O atoms is more acidic; e.g., sulfuric acid versus sulfurous acid.
- Another way of saying it: If the oxidation number increases, the acidity increases.



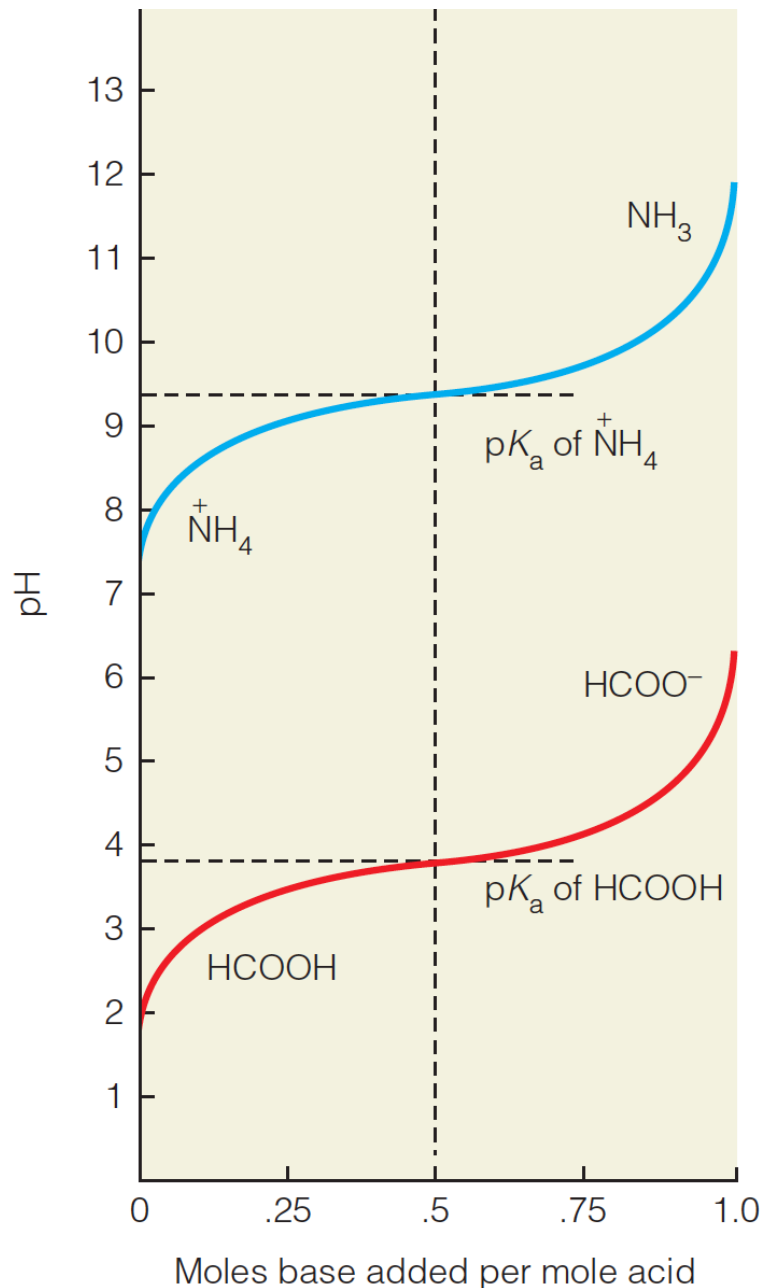
Carboxylic Acids

- **Carboxylic acids** are organic acids containing the —COOH group.
- Factors contributing to their acidic behavior:
 - Other O attached to C draws electron density from O—H bond, increasing polarity.
 - Its conjugate base (carboxylate anion) has resonance forms to stabilize the anion.



The **Henderson–Hasselbalch equation** describes the change in pH during titration of a weak acid or a weak base.

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$



Ionic Equilibria

Buffered solutions are able to minimize the change in pH following addition of acid or base because the conjugate acid (HA) and conjugate base (A^-) of the buffering compound **are present in sufficient** concentration to combine with the added H^+ or OH^- and neutralize them.

Buffer solutions function because the pH of a weak acid–base solution is least sensitive to added acid or base **near the pK_a** , *where the* conjugate acid and conjugate base of the buffer are both present in nearly ***equimolar concentrations***.

Ionic Equilibria

- Organisms must maintain the pH inside cells and in most bodily fluids within the narrow pH range of about 6.5 to 8.0.
- The dihydrogen phosphate–hydrogen phosphate system, with a pK_a of 6.86, plays a major role in controlling *intracellular* pH because phosphate is abundant in cells.
- **Blood** contains dissolved CO_2 as a waste product of metabolism, therefore the carbonic acid–bicarbonate system provides considerable buffering capacity.